Lecture 16: The Pigeonhole Principle

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1 Socks in a Drawer

Your drawer contains red and black socks, with plenty of each color. You draw socks without looking. How many must you draw to **guarantee two of the same color**?

Two do not suffice: you might draw one red sock and one black sock. Three suffice. If the first two already match, we are done. Otherwise, the third sock must match one of the first two. Thus the answer is

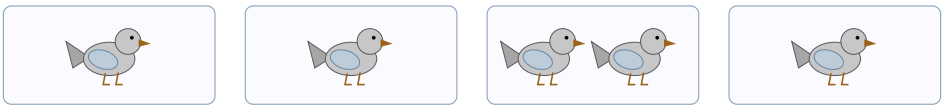
$$\boxed{\text{3 socks.}}$$

The word *guarantee* matters. We want a conclusion that holds for every possible draw, including the least favorable one.

2 The Pigeonhole Principle

Theorem 1 (Pigeonhole principle)**.** *If more than n objects are placed into n boxes, where $n \geq 1$, then at least one box contains at least two objects.*

The traditional picture uses pigeons and pigeonholes. In the example below, five pigeons occupy four holes; at least one hole must be shared.



Hole 1 Hole 2 Hole 3 Hole 4

Proof. Suppose every box contained at most one object. Then the n boxes would contain at most n objects altogether. This contradicts the assumption that there are more than n objects. □

For the sock problem, the three socks are the objects and the two colors are the boxes. Place each sock in the box corresponding to its color. Two socks in one box have the same color.

The useful boxes need not be physical containers. They may be colors, remainders, regions, or pairs of vertices. In each application, ask: *What are the objects? What are the boxes? What does sharing a box imply?*

A useful extension. If N objects are placed into k boxes, some box contains at least $\lceil N/k \rceil$ objects. Otherwise every box would contain fewer than N/k objects, so the total would be less than N.

3 Two Numbers That Sum to Eleven

Problem. Choose any six *distinct* numbers from $\{1, 2, \ldots, 10\}$. Prove that two of the chosen numbers sum to 11.

Proof. Partition the ten numbers into five pairs:

$\{1, 10\}$ $\{2, 9\}$ $\{3, 8\}$ $\{4, 7\}$ $\{5, 6\}$

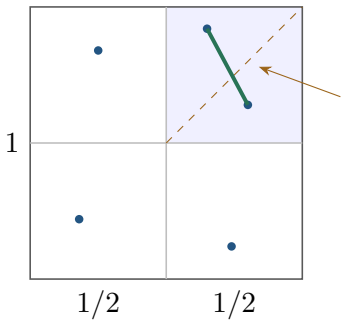
Use these pairs as boxes, and put each chosen number into its pair. There are six chosen numbers and only five boxes, so some box contains two chosen numbers. Since the numbers are distinct, both members of that pair were chosen. Their sum is 11. □

Pigeons: the six chosen numbers. **Holes:** the five displayed pairs.

The bound is sharp: choosing $\{1, 2, 3, 4, 5\}$ gives five numbers with no pair summing to 11.

4 Five Points in a Unit Square

Problem. Five points lie in a square of side length 1, possibly on its boundary. Prove that two of them are at distance at most $\sqrt{2}/2$.



Small-square diagonal:
$\sqrt{(1/2)^2 + (1/2)^2} = \sqrt{2}/2$

Proof. Divide the square into four smaller squares of side length $1/2$. Assign a point on the vertical dividing line to the right half, and a point on the horizontal dividing line to the upper half. Thus each point belongs to exactly one region.

Five points occupy four regions, so two points belong to the same region. Their horizontal and vertical separations are each at most $1/2$. By the Pythagorean theorem, their distance is at most

$$\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{2}}{2}.$$

□

Pigeons: the five points. **Holes:** the four regions.

5 A Multiple of 2003 Made Entirely of Sevens

Problem. Prove that *some term* of the sequence

$$7, \quad 77, \quad 777, \quad 7777, \quad \ldots$$

is divisible by 2003.

Write

$$a_k = \underbrace{77\cdots7}_{k \text{ digits}}.$$

We will prove that one of the first 2003 terms works.

Proof. Consider $a_1, a_2, \ldots, a_{2003}$. If one of them has remainder 0 when divided by 2003, we are done.

Otherwise their remainders all lie in

$$\{1, 2, \ldots, 2002\}.$$

There are 2003 terms and only 2002 possible nonzero remainders. By the pigeonhole principle, two terms have the same remainder. Choose their indices so that $i < j$. Then

$$a_i \equiv a_j \pmod{2003}, \qquad \text{so} \qquad 2003 \mid (a_j - a_i).$$

Subtracting the shorter string of sevens from the longer one leaves a string of sevens followed by zeros. For example,

$$777777 - 777 = 777000 = 777 \cdot 10^3.$$

In general,

$$a_j - a_i = \underbrace{77\cdots7}_{j-i \text{ sevens}}\,\underbrace{00\cdots0}_{i \text{ zeros}} = a_{j-i}\,10^i.$$

Thus $2003 \mid a_{j-i}10^i$. Since 2003 is divisible by neither 2 nor 5, we have $\gcd(2003, 10^i) = 1$. The coprime cancellation property therefore gives

$$\boxed{2003 \mid a_{j-i}.}$$

Because $j - i \geq 1$, this is a term of the original sequence. □

Pigeons: the first 2003 terms. **Holes:** the 2002 nonzero remainders, assuming no term has remainder zero.

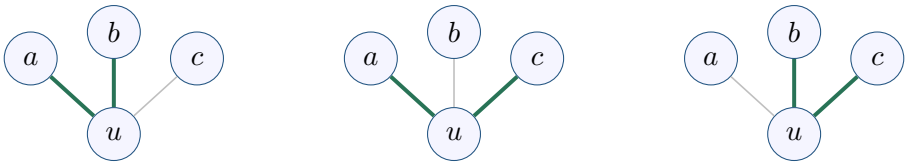
What made the argument work? Equal remainders give a divisible *difference*. The special form of these numbers turns that difference into another term times a power of 10. Coprimality lets us remove the power of 10. We have proved existence without computing the required term.

6 A Four Cycle Forced by Counting

Problem. Let G be a simple undirected graph with six vertices, each of degree exactly 3. Prove that G contains a cycle of length 4.

Step 1: Count paths with two edges. Fix a vertex u. Its three distinct neighbors can be called a, b, c. There are exactly three two-edge paths with u as their center:

$$a - u - b, \qquad a - u - c, \qquad b - u - c.$$



$a - u - b$ $\qquad$ $a - u - c$ $\qquad$ $b - u - c$

The corresponding endpoint pairs are $\{a, b\}$, $\{a, c\}$, and $\{b, c\}$. We count a path and its reversal only once. Each of the six vertices contributes three paths. Every such path has a unique center, so the total is $6 \cdot 3 = \boxed{18}$.

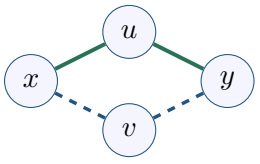
Step 2: List the possible endpoint pairs. Label all six vertices $1, 2, 3, 4, 5, 6$. Every unordered pair appears exactly once in the following list:

$$\begin{array}{ccccc} \{1,2\} & \{1,3\} & \{1,4\} & \{1,5\} & \{1,6\} \\ & \{2,3\} & \{2,4\} & \{2,5\} & \{2,6\} \\ & & \{3,4\} & \{3,5\} & \{3,6\} \\ & & & \{4,5\} & \{4,6\} \\ & & & & \{5,6\} \end{array}$$

There are $5 + 4 + 3 + 2 + 1 = \boxed{15}$ pairs. Later, we will write this count as $\binom{6}{2} = 6 \cdot 5/2$ and learn the general formula. The list above is all we need here.

Step 3: Apply pigeonhole. Put each of the 18 paths into the box labeled by its endpoint pair. There are only 15 boxes, so two distinct paths have the same endpoints, say x and y.

Their centers must be different: for a fixed center and endpoint pair, there is only one such path. Call the centers u and v. All four vertices are distinct, and the paths $x - u - y$ and $x - v - y$ give the cycle



$\boxed{x - u - y - v - x.}$

Pigeons: the 18 two-edge paths. **Holes:** the 15 endpoint pairs.

7 An Even Length Path in a Tree

Problem. Choose any three distinct vertices of a tree. Prove that two of them are joined by a path containing an even number of edges.

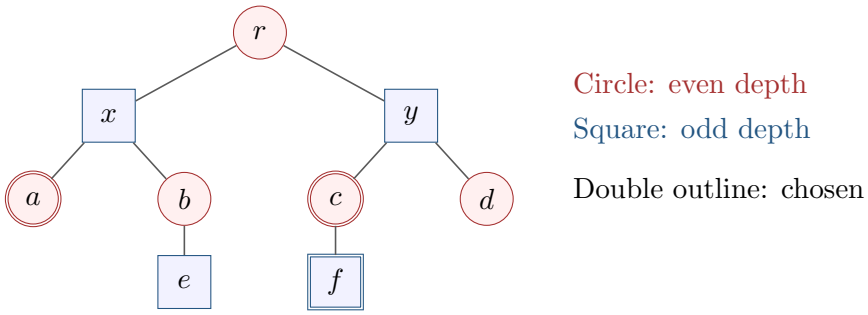
Recall that the **length** of a path is its number of edges. The **degree** of a vertex is its number of incident edges. The claim concerns an even *length*.

Proof. Choose any vertex r as a root. Give each vertex one of two colors:

- red if its distance from r is even;
- blue if its distance from r is odd.

Every edge of a rooted tree joins a parent to a child, so its endpoints have distances from r that differ by exactly one. Hence every edge joins opposite colors.

The three chosen vertices occupy two color classes. By pigeonhole, two of them have the same color. Follow the unique path between them. The color changes at each edge. After an odd number of edges the color is different; after an even number it is the same. Because the two endpoints have the same color, this path has even length. □



In this example the chosen vertices are a, c, f. Both a and c are red, and their path $a - x - r - y - c$ has four edges.

Pigeons: the three chosen vertices. **Holes:** the two color classes.

Path length and distance. In a tree, the unique path between two vertices has length equal to their distance. In a general graph these ideas must be distinguished: two vertices of a triangle are at distance 1, even though a path of length 2 also joins them.

Extension. Among any $2k + 1$ chosen vertices of a tree, at least $k + 1$ have the same color. Every pair among those $k + 1$ vertices is therefore at even distance.

8 Friends and Strangers

Theorem 2 (The six person Ramsey theorem)**.** *Among any six people, there are three mutual friends or three mutual strangers. Friendship is assumed to be mutual.*

Three **mutual friends** means that each person is friends with both others. Three **mutual strangers** means that no two of the three are friends.

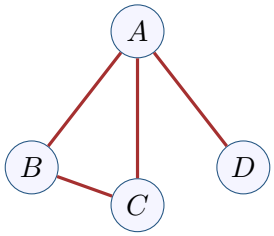
8.1 Translate the Claim into a Graph

Draw a vertex for each person and join every pair of vertices. Color an edge red when the two people are friends and blue when they are strangers. We must find a **monochromatic triangle**: a triangle whose three edges all have the same color. In the figures, red edges are solid and blue edges are dashed.

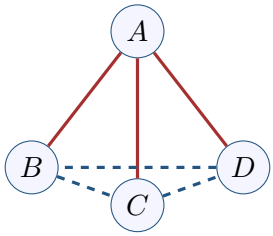
Proof. Choose one person A. The five edges from A have two possible colors. By pigeonhole, at least three have the same color: otherwise there would be at most two of each color, only four edges. Call their other endpoints B, C, D.

First suppose AB, AC, AD are all red. Examine BC, BD, CD.

- If any one is red, say BC, then AB, AC, BC form a red triangle. Thus A, B, C are mutual friends.
- If none is red, all three are blue. Then B, C, D are mutual strangers.



A red edge BC gives the red triangle ABC.



No red edge among B, C, D gives the blue triangle BCD.

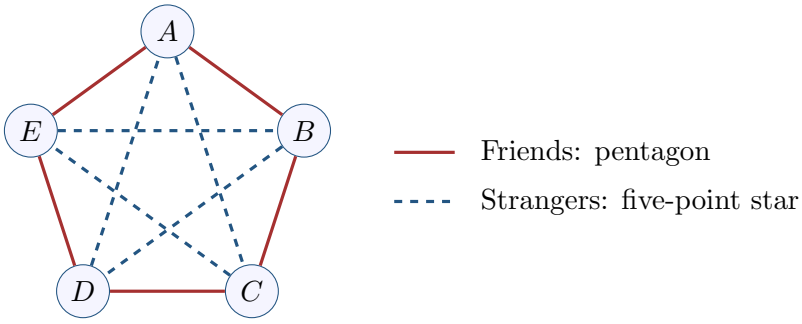
If AB, AC, AD are all blue, exchange the roles of the colors. A blue edge among B, C, D completes a blue triangle with A; if there is no such edge, B, C, D form a red triangle. Every case gives the required trio. □

Pigeons: the five edges incident to A. **Holes:** red and blue.

The two parts of the proof. Pigeonhole gives three edges of one color from A. Examining the edges among their other endpoints then gives a triangle. Both parts are needed: three people who are all friends with A need not be friends with one another.

8.2 Why Six Is the Smallest Number

Five people need not contain either kind of trio. Arrange them around a pentagon, with each person friends with exactly their two neighbors around the pentagon.



The red friendship edges form the five-cycle $A - B - C - D - E - A$, so there is no red triangle. The blue stranger edges form the five-cycle $A - C - E - B - D - A$, so there is no blue triangle either. Crossings in the drawing are not additional vertices.

We have shown that six people always suffice and that five may fail. The notation for this exact threshold is

$$\boxed{R(3,3) = 6.}$$

Here $R(3,3)$ is the smallest number of people that guarantees either three mutual friends or three mutual strangers.